

Prediction of Mechanical Properties of Heat-treated EN AW-6082 using Symbolic Regression

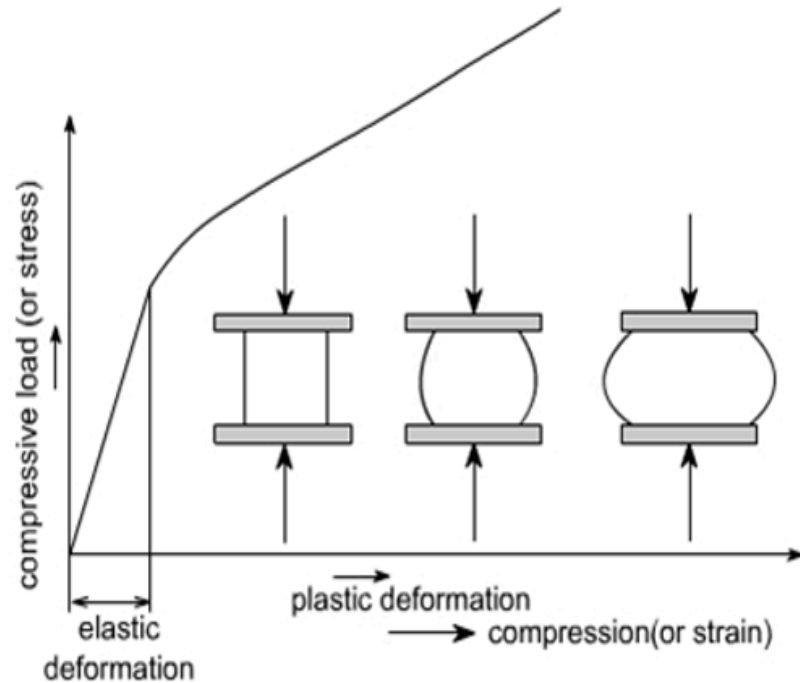
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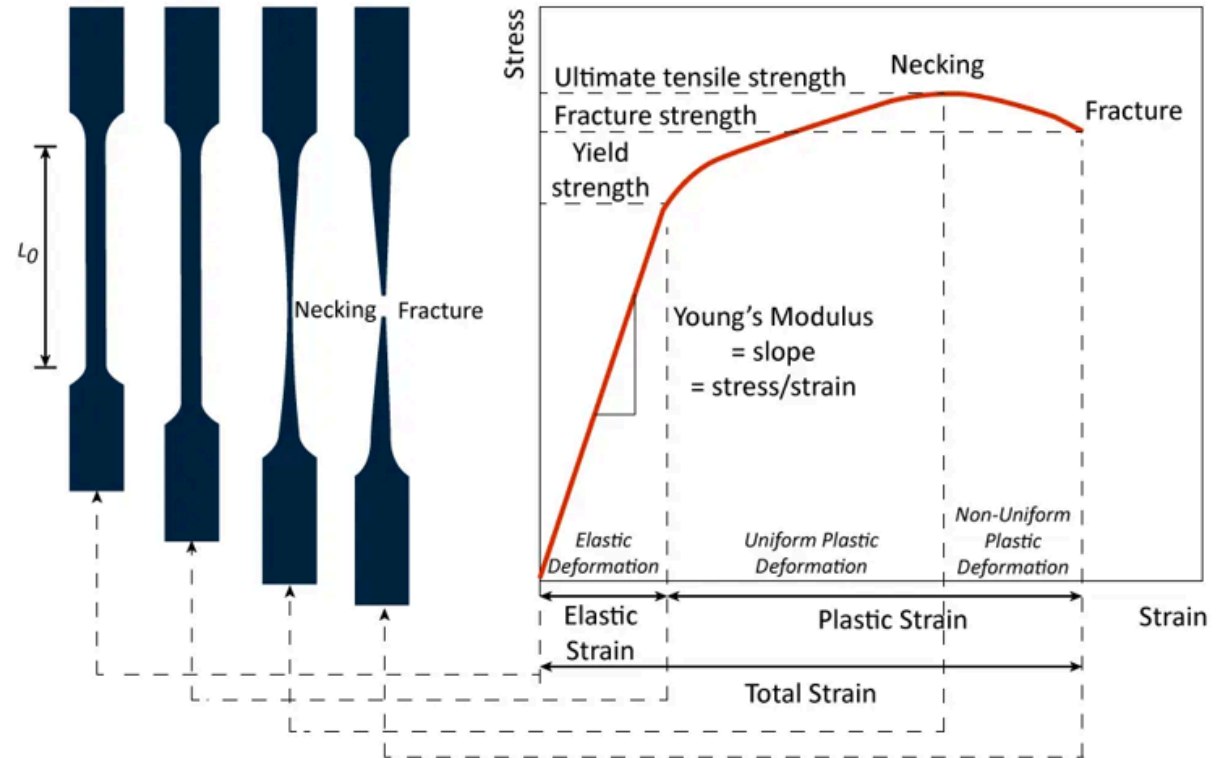
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Mechanical Testing and Stress-Strain Curves



Virtual Labs



Effect of Specimen Geometry on Tensile Testing Results - ADMET

Heat-treatment of Aluminum Alloys

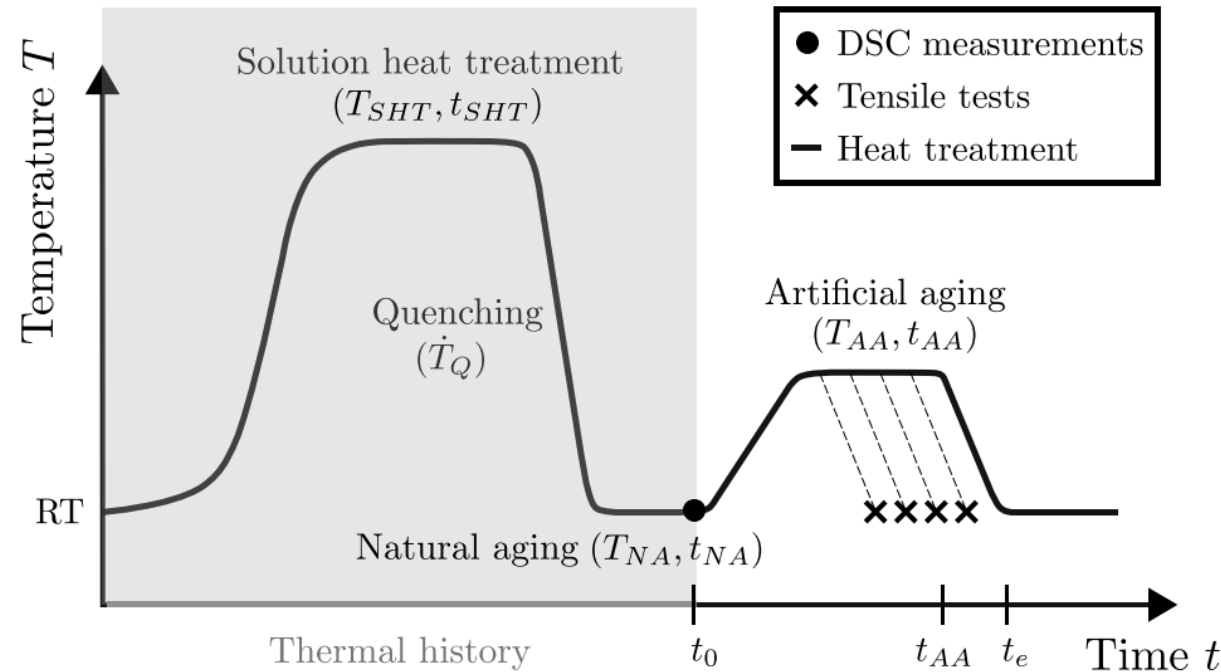


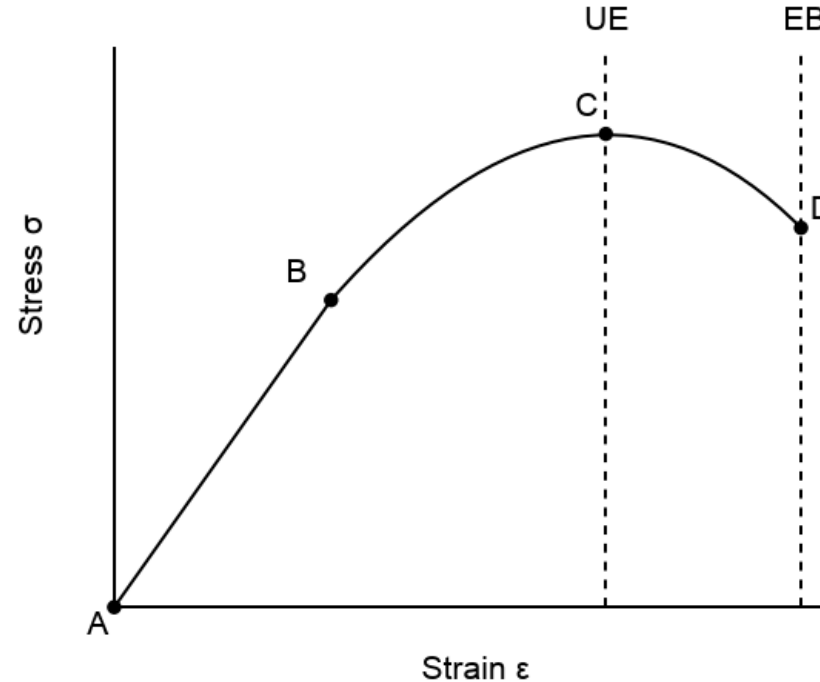
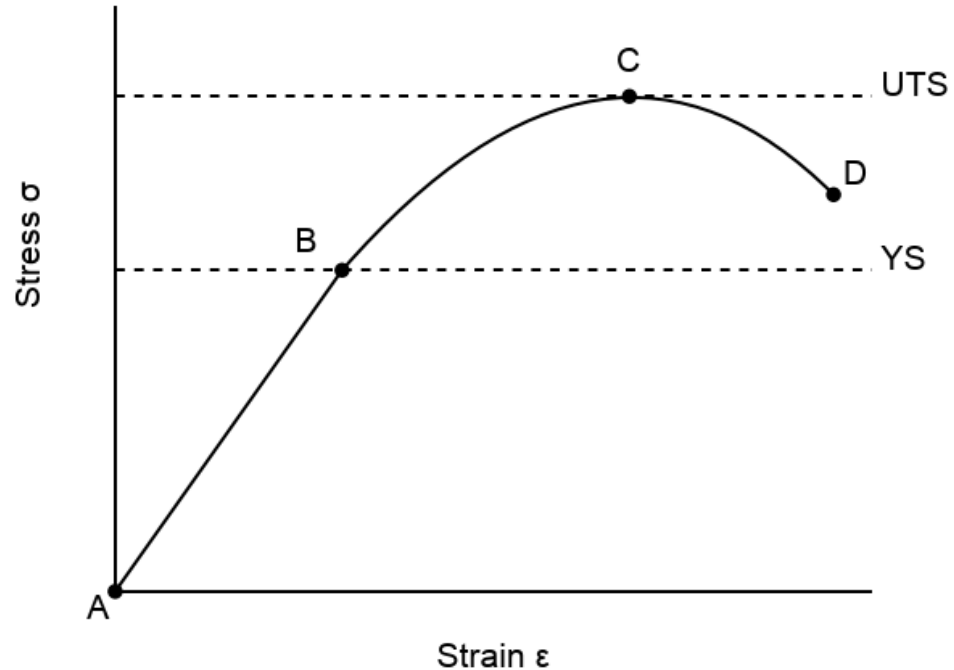
Fig. 1: Typical heat treatment of an aluminum alloy and measurements carried out within this study.

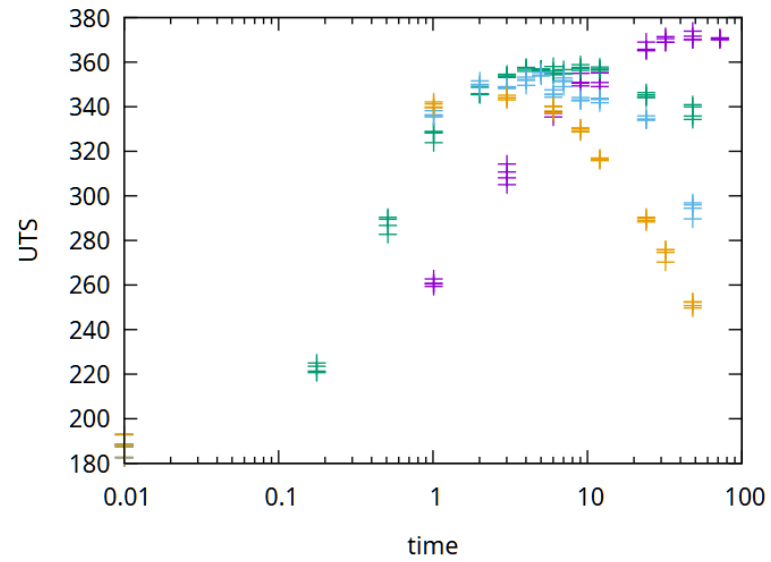
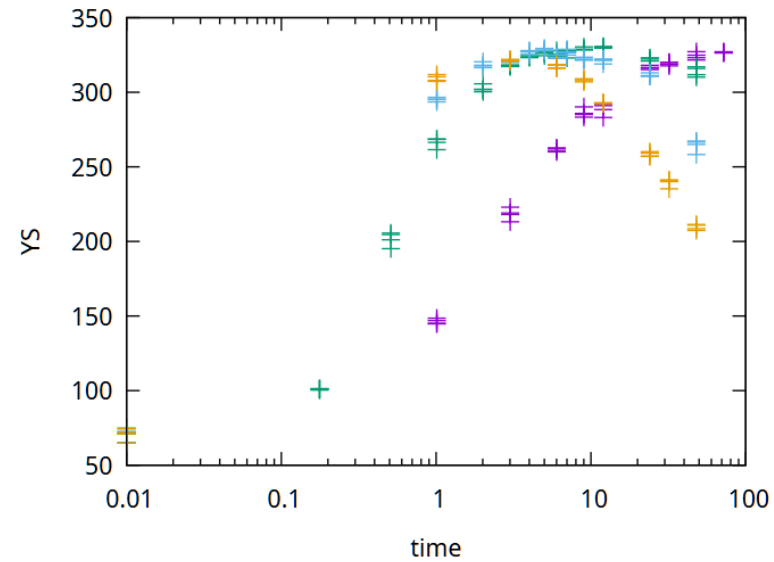
L. Grohmann et al., "Age-hardening model for Al-Mg-Si alloys: a computationally efficient multi-phase approach to predict the evolution of yield strength", Mathematical and Computer Modelling of Dynamical Systems, 2026

Objective: Empirical Modeling of Mechanical Properties

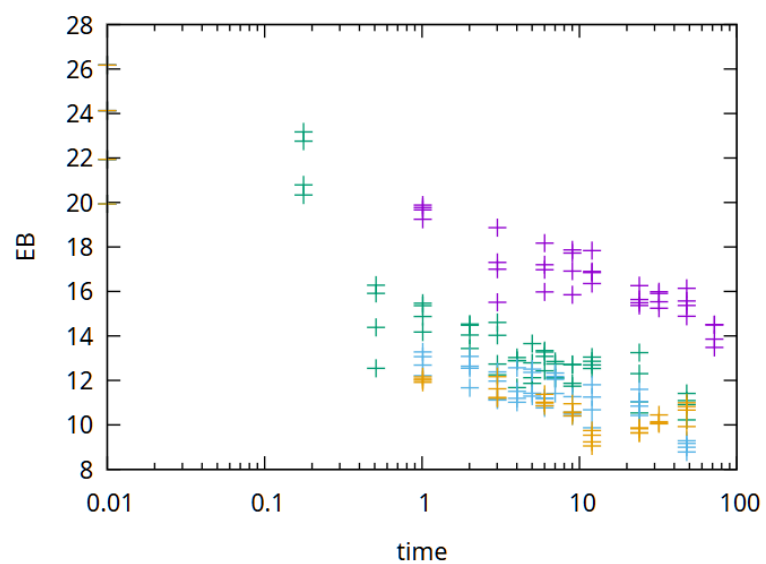
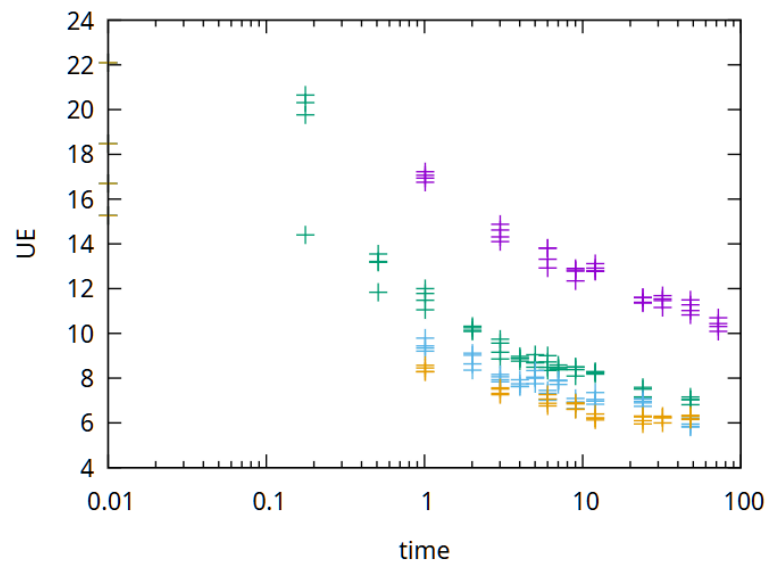
Predict four mechanical properties as a function of aging temperature (T_{AA}) and duration (t_{AA}) :

- Ultimate tensile strength (UTS), yield strength (YS)
- Uniform elongation (UE), and elongation at break (EB).



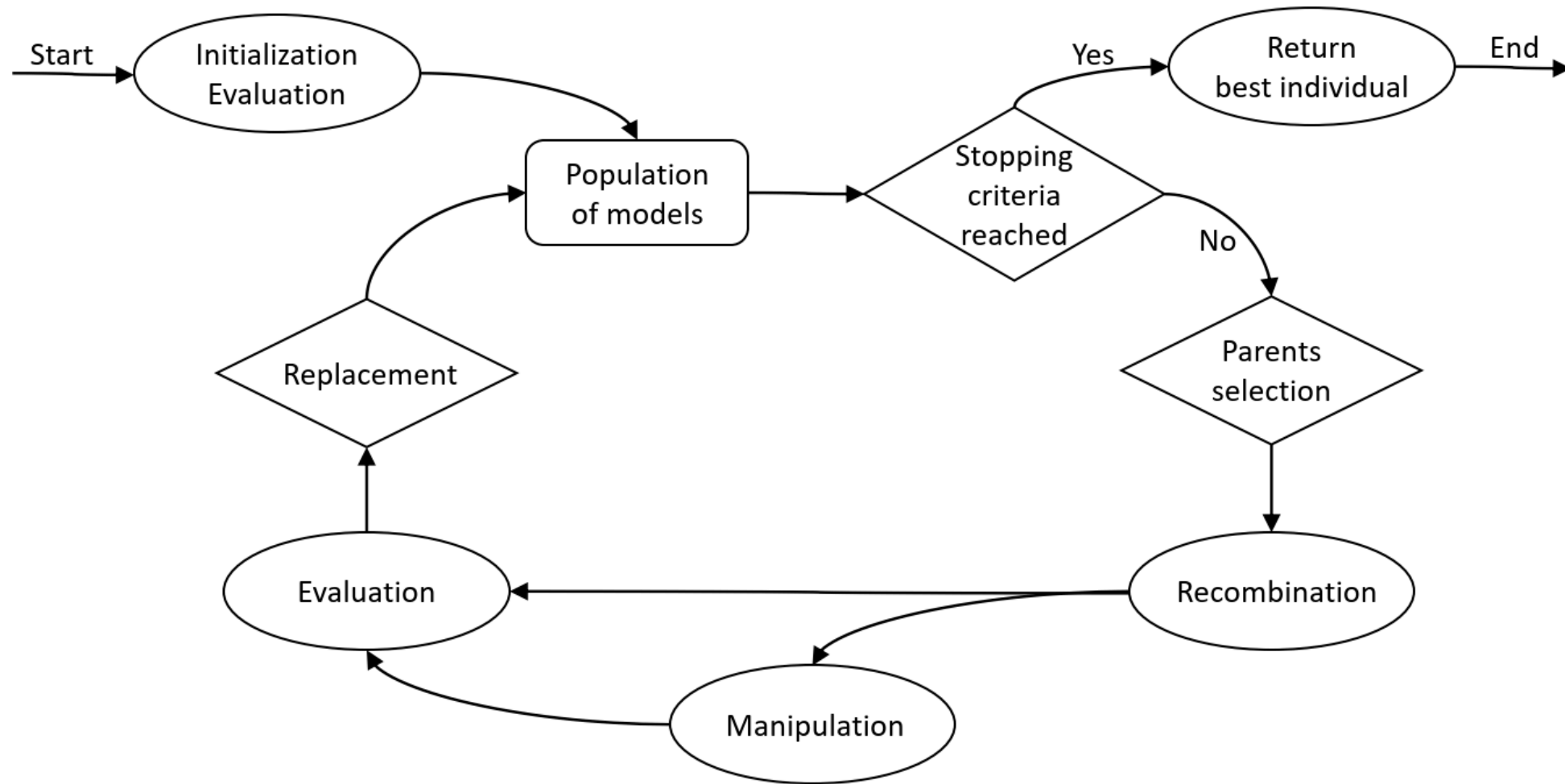


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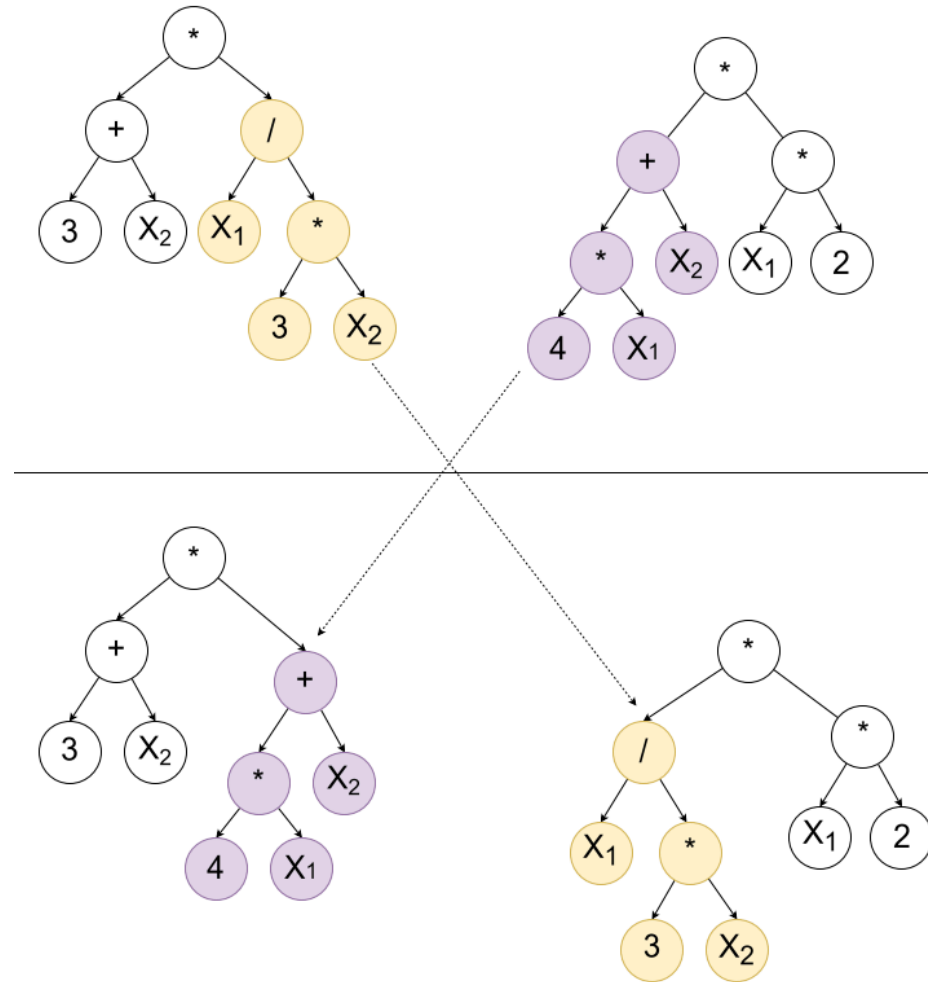


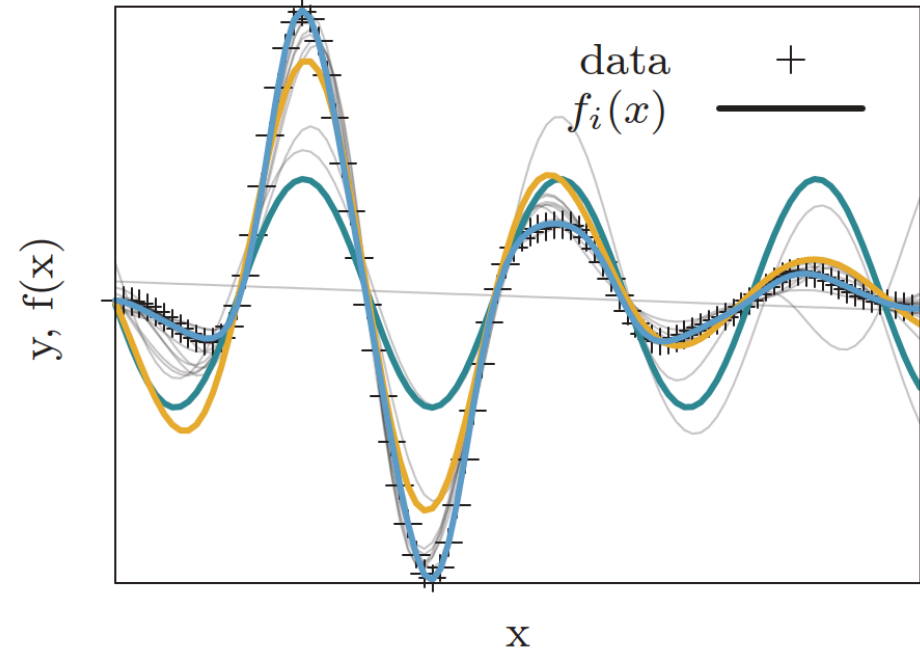
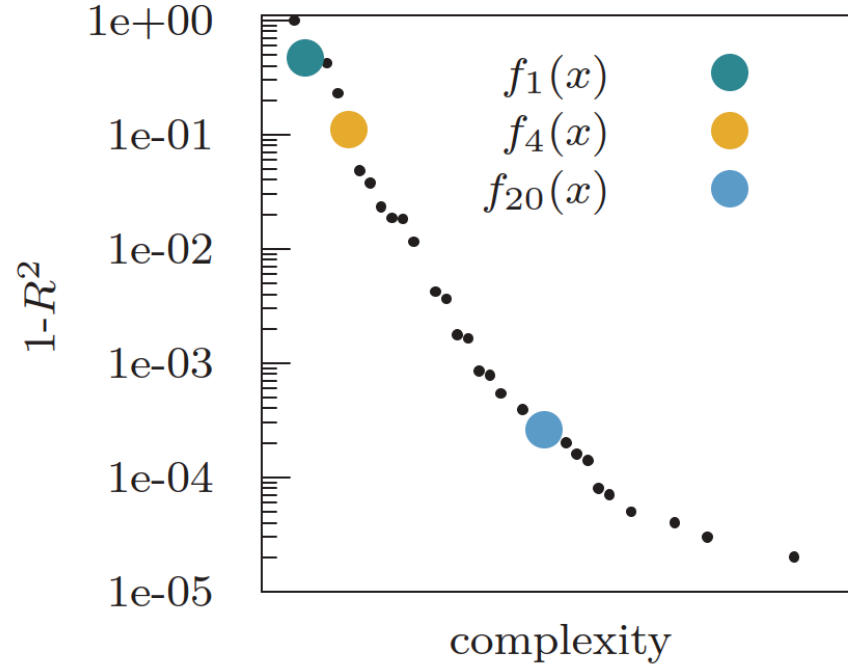
data (temp=140) +
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Evolutionary Algorithms and Genetic Programming



Expression Tree Encoding and Subtree Crossover in Symbolic Regression





$$f_1(x) = \sin(1.992 x)$$

$$f_4(x) = \exp(\sin(0.526 x)) \sin(1.995 x)$$

$$f_{20}(x) = \sin(0.307 x) \sin(2 x)$$

$$\exp \left(\sin(0.637 x) - \cos(\sin(\sin(0.485 x)) - 1.341) \right.$$

$$\left. \cos(0.461 x)(\sin(2.082 x) + 0.123 x) \right)$$

Information theory: study of communication of information through a channel.

The **description length (DL)** is the total length of the message required to communicate a dataset.

$$DL = L(D|M) + L(M)$$

Application of Occham's razor.

- The MDL model allows to compress D well, and is itself short.
- If M predicts y well, we only need to encode the residuals, which allows a shorter message.
- D are the y values; it is assumed that the reciever has X .

Description length for symbolic regression (Bartlett et al. 2023):

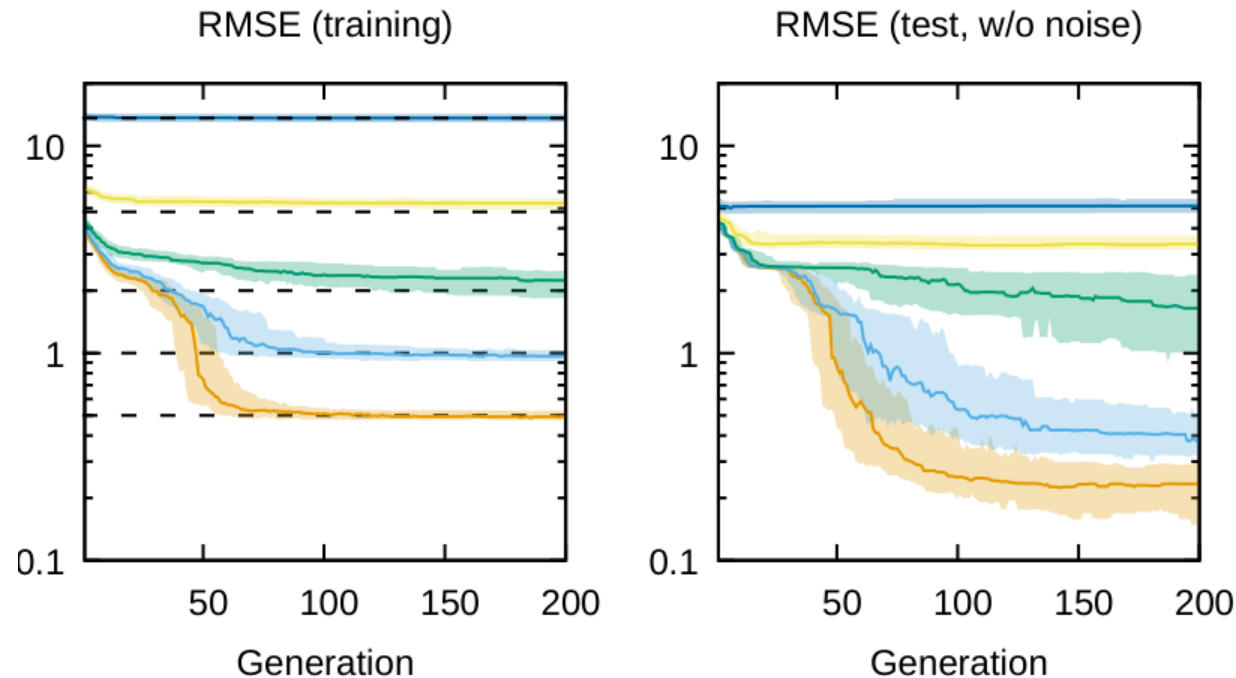
$$\begin{aligned} & -\log(\mathcal{L}(\hat{\boldsymbol{\theta}})) + k \log(n) + p \log(2) + \sum_j \log(c_j) \\ & + \sum_i^p \log(|\hat{\theta}_i|/\Delta_i), \end{aligned}$$

- The log-likelihood term prefers models that fit the data well (small residuals)
- The $k \log n$ term penalizes program length k with n distinct symbols
- $\log(c_j)$ penalizes large constants
- $\log |\hat{\theta}_i|$ penalizes parameters of large magnitude
- Only the quantization grid Δ_i for the MLE ($\hat{\theta}$) is a free parameter but can be calculated from the Hessian of the log-likelihood.

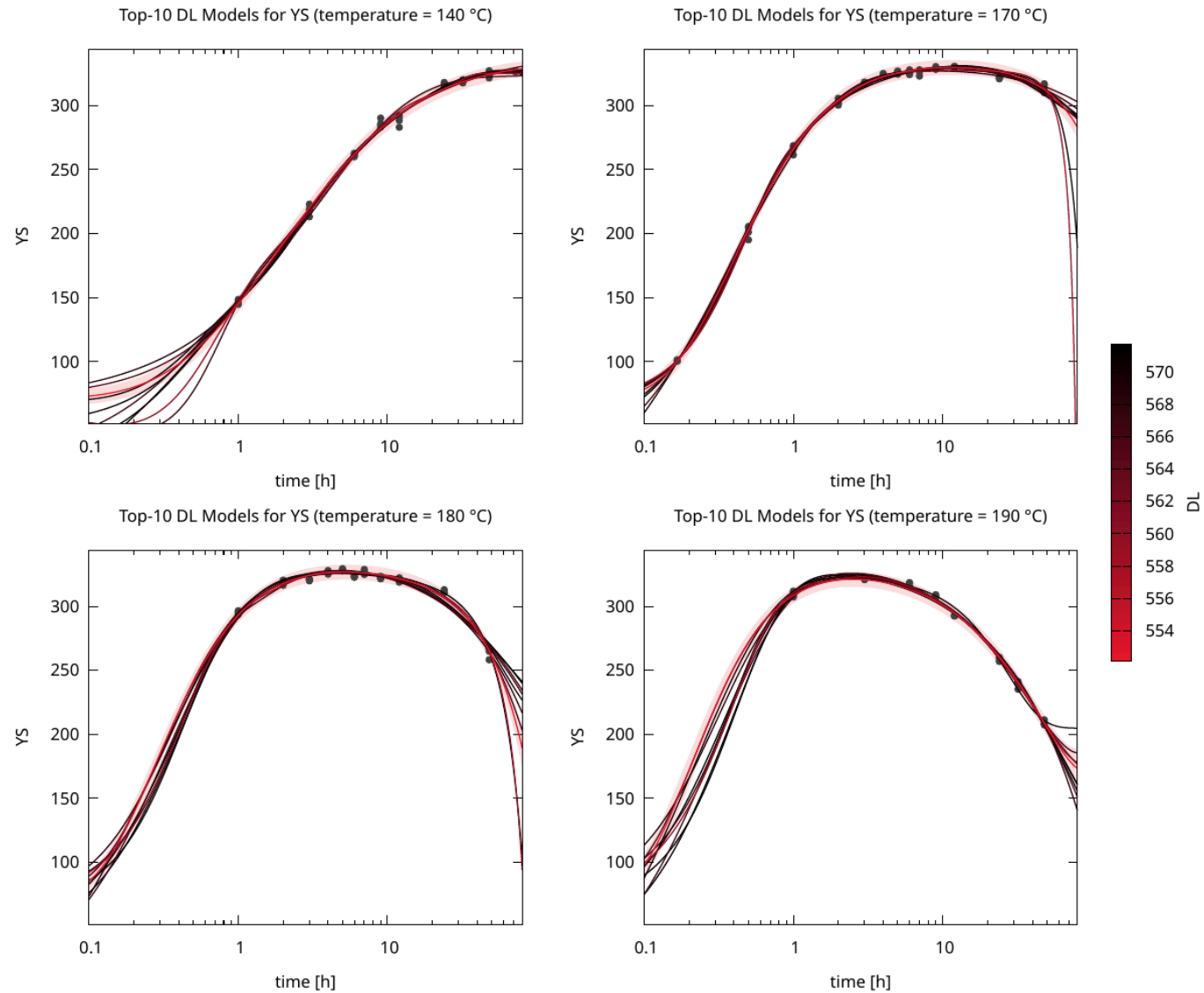
D. Bartlett, H. Desmond, P.G. Ferreira: Exhaustive Symbolic Regression, 2023

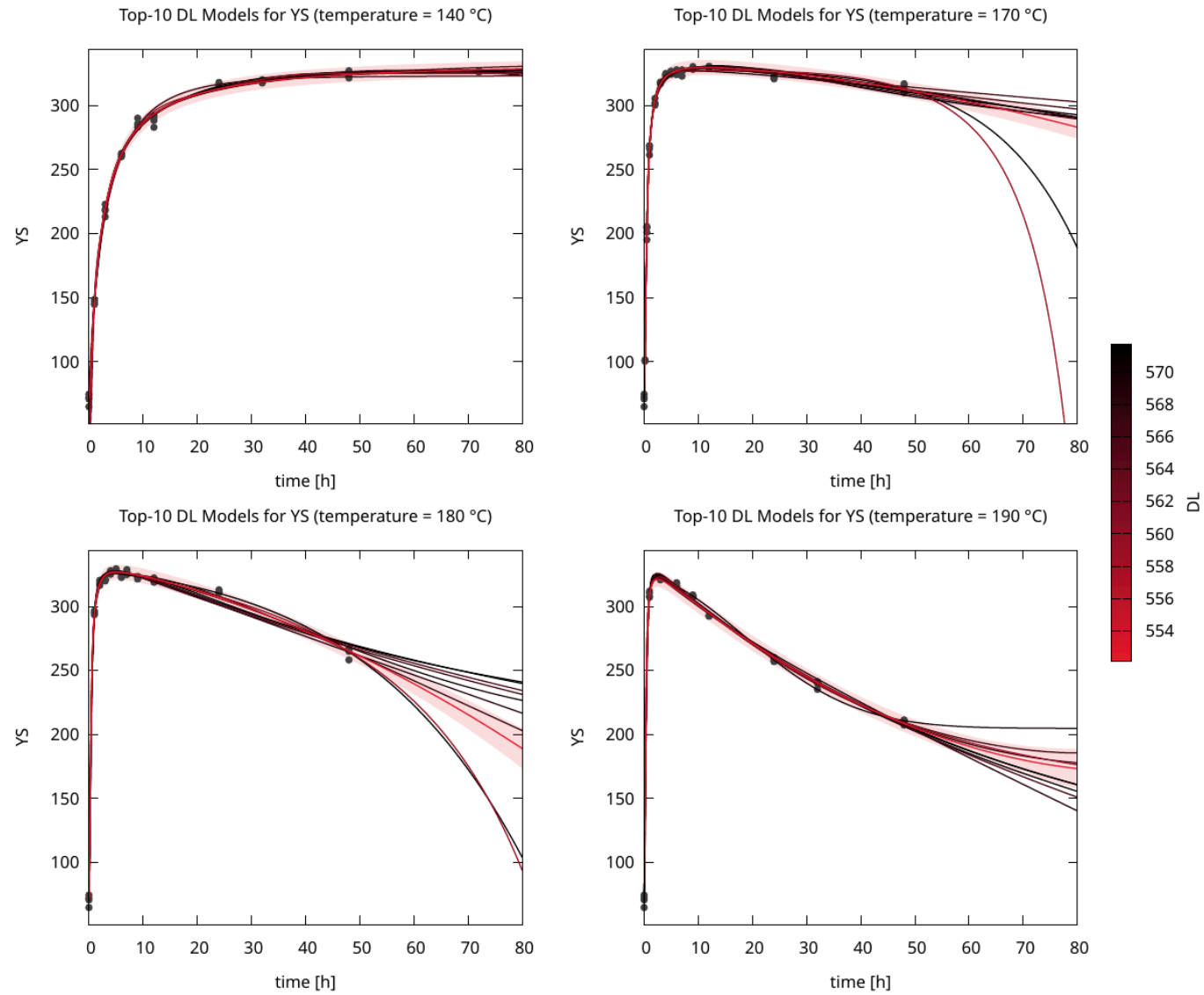
MDL allows us to automatically infer the best model complexity for a given dataset.
No overfitting. No cross-validation necessary.

Example: 5 different noise levels, 100 training points, 100 GP runs per noise level.



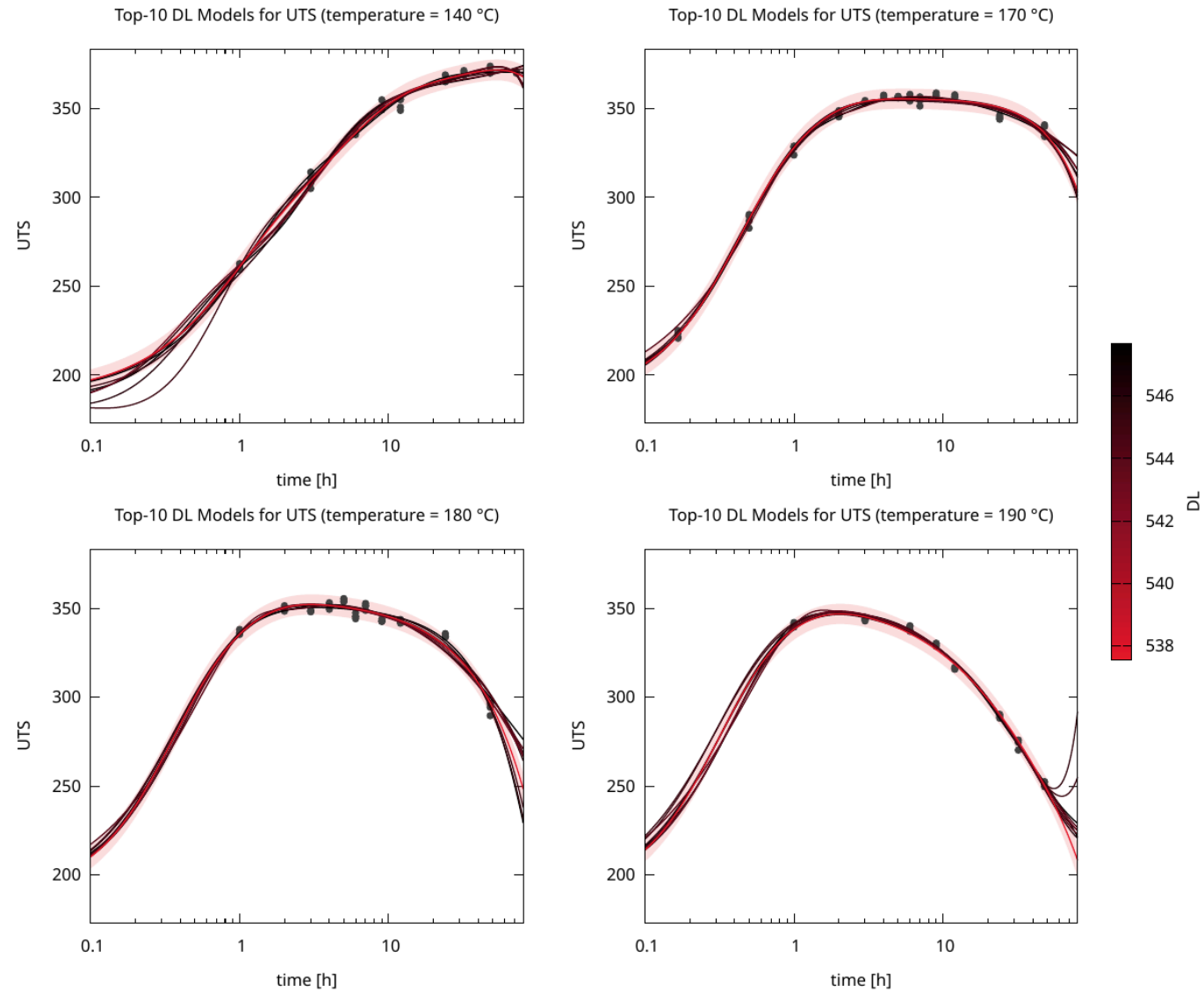
G. Kronberger et al., 2026, Guiding Multi-Objective Genetic Programming with Description Length Improves Symbolic Regression Solutions", arxiv:2605.22374

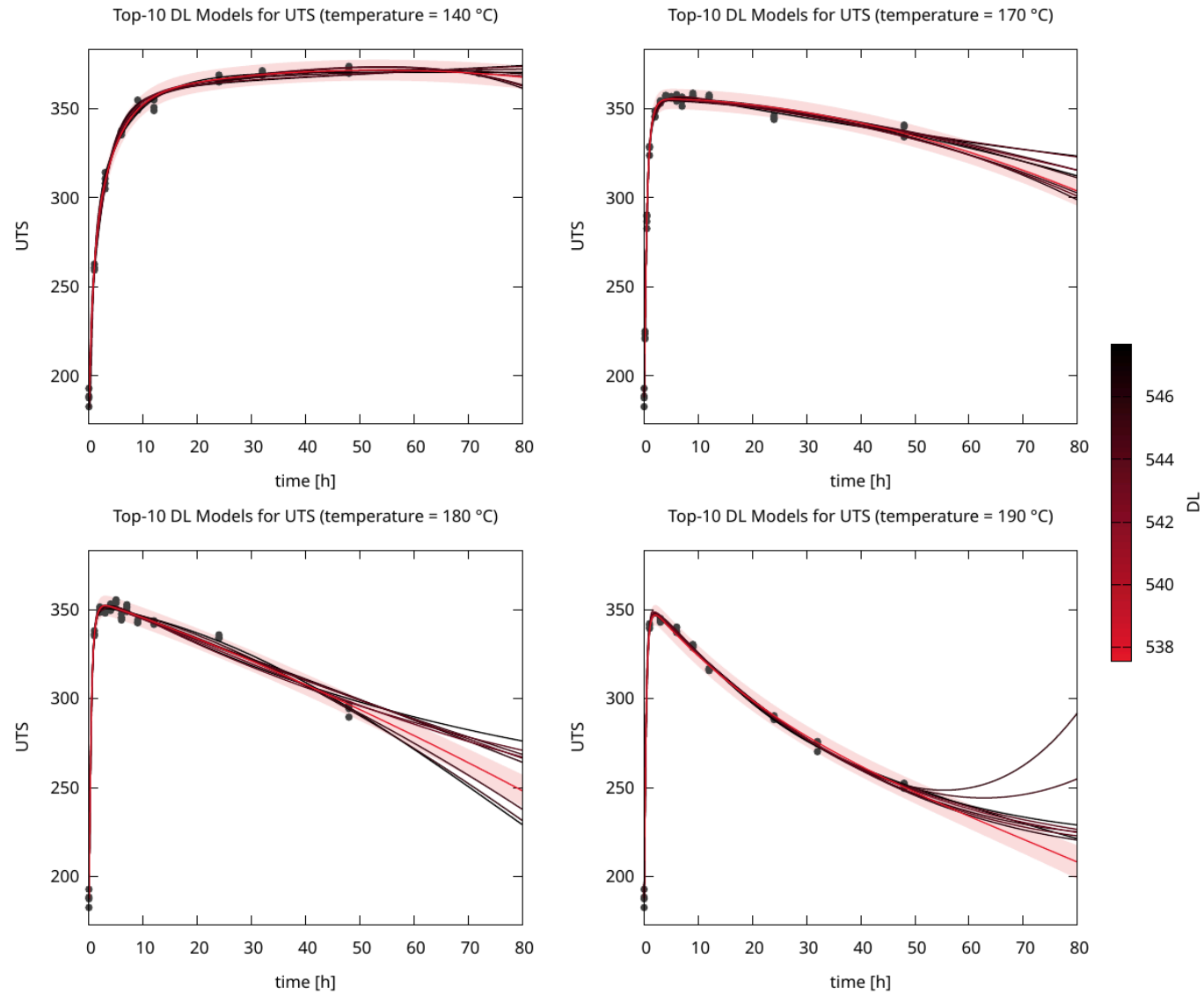


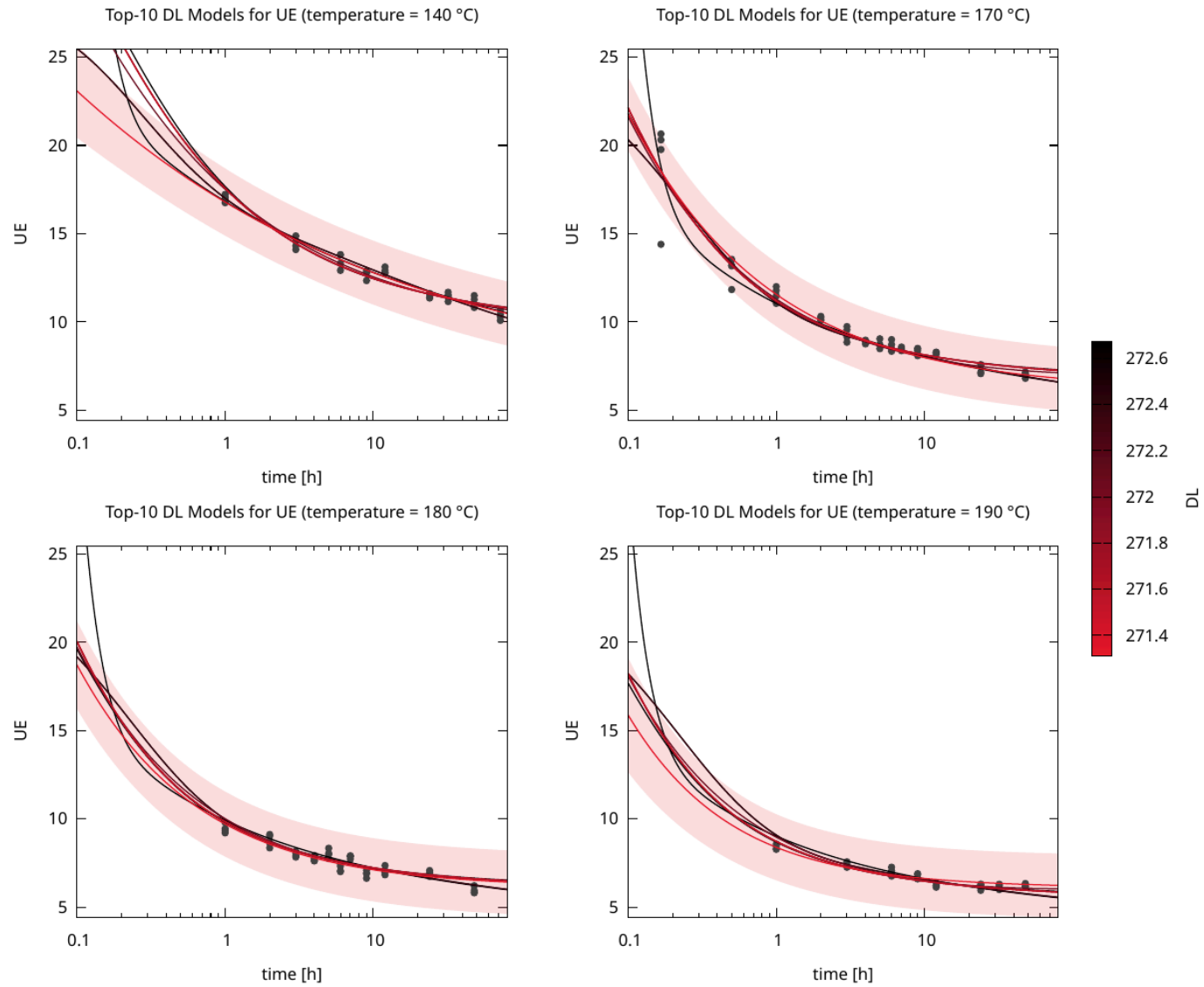


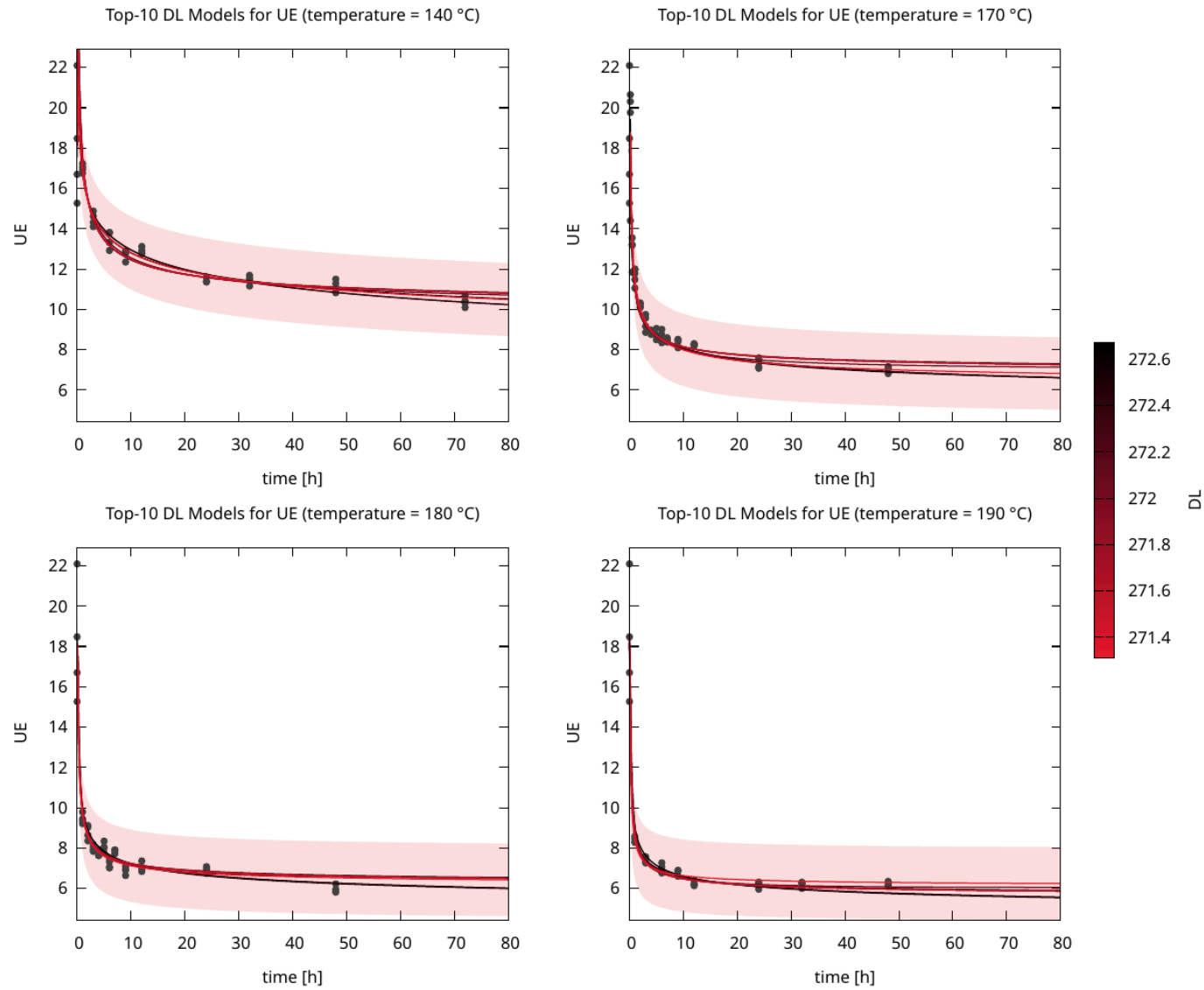
YS Expression

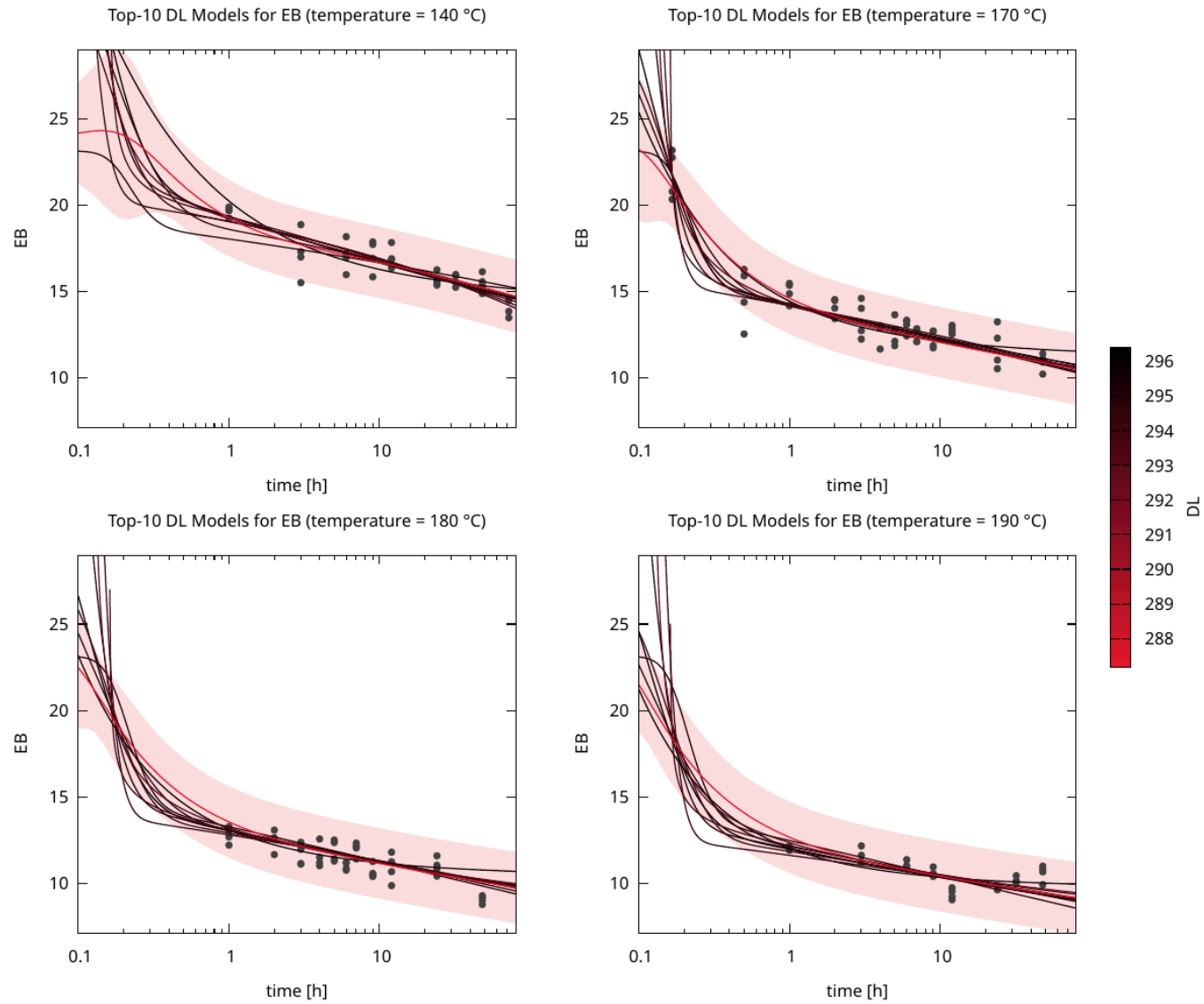
$$\text{YS}(t, T) = e^{T\theta_1} e^{t^{\theta_2} (T\theta_3 + \theta_4)} \theta_5 + \theta_6 - t (T\theta_7)^{T\theta_8 + \theta_9} (t (T\theta_{10} + \theta_{11}) + \theta_{12}) + \theta_{13}$$

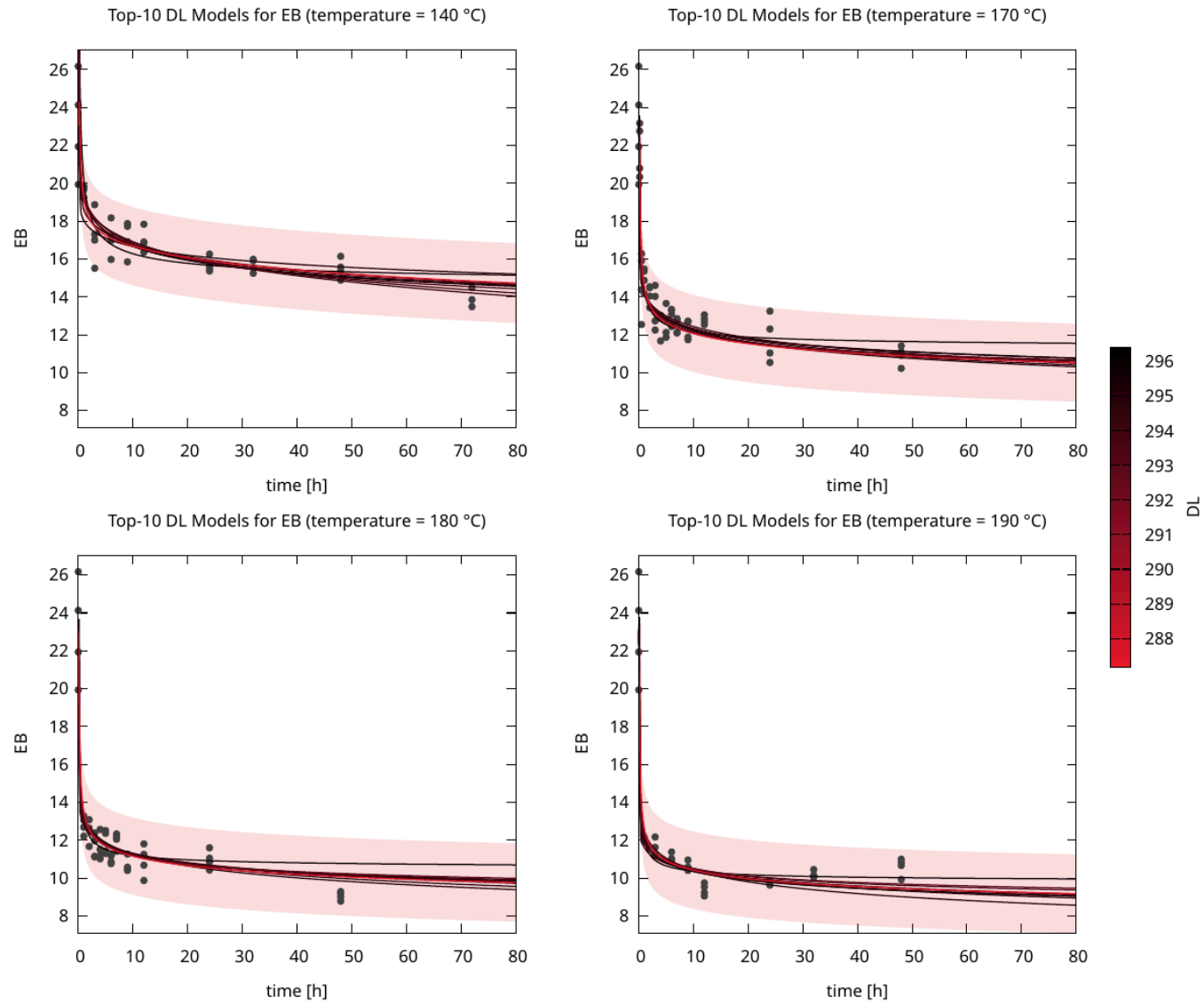












Result overview

Target	Best model (DL)	RMSE (train)	# Param	Length
YS	552.1	2.91 [MPa]	13	43
UTS	537.6	2.85 [MPa]	13	44
UE	271.3	0.91 [%]	7	7
EB	287.2	1.03 [%]	7	7

Conclusions

- SR can be used to produce short and accurate formulas for predicting mechanical properties of heat-treated aluminum alloys.
- Using DL for fitness in GPSR automatically selects models of appropriate complexity.
- The models provide smooth predictions of mechanical properties, which may be useful for process optimization of the aging process.

Limitations

- The resulting models are empirical only and cannot be interpreted in terms of the underlying physics of the aging process.
- Only a single alloy (EN AW-6082) was considered. The models are not applicable to other alloys.

Questions

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Tensile tests

Tensile tests were performed in SSSS (also called W temper) and after artificial aging at different aging temperatures for different aging times.

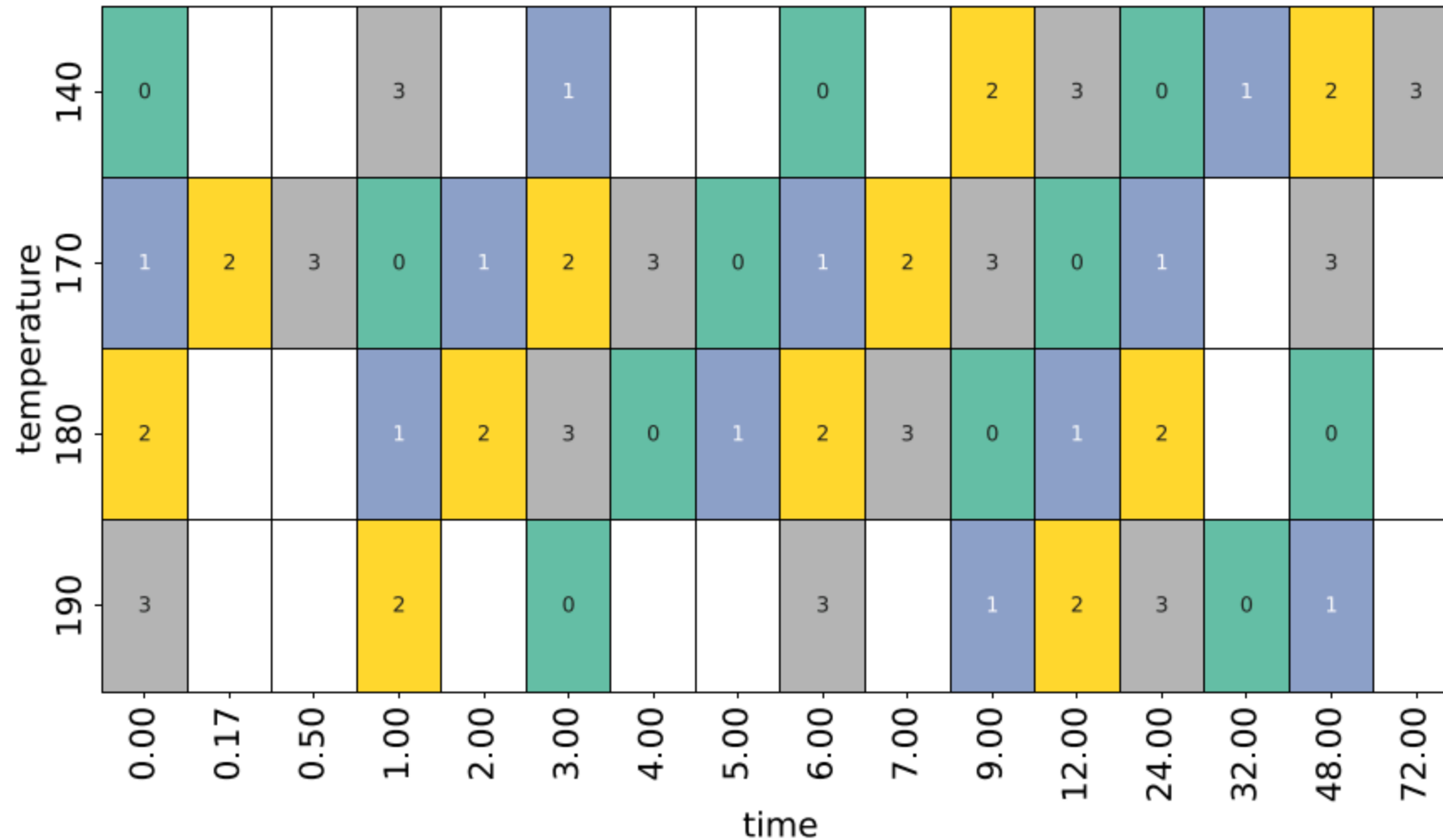
The measurements were carried out using a Zwick 250 kN testing machine.

All samples had a reduced section width of 12.5 mm, a gauge length of 50 mm, and a reduced section length of 75 mm (DIN 50125).

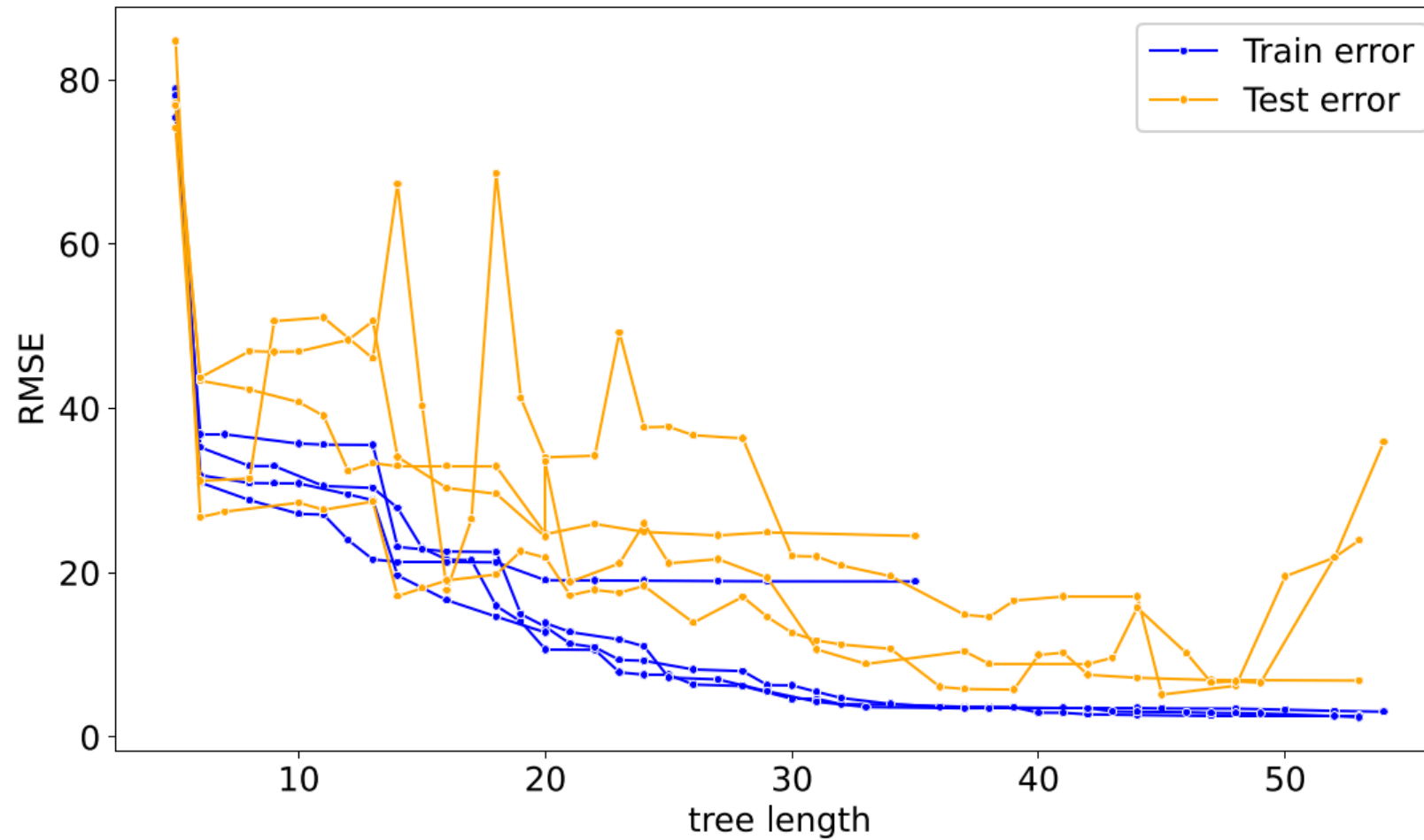
Measurements were carried out 4x (ONORM-EN-ISO 6892-1 standard).

The 0.2% proof stress (yield strength) was calculated based on the averaged stress-strain curves.

Systematic 4x Cross-validation Fold Assignment



Cross-validation for the YS model



Cross-validation for the YS model

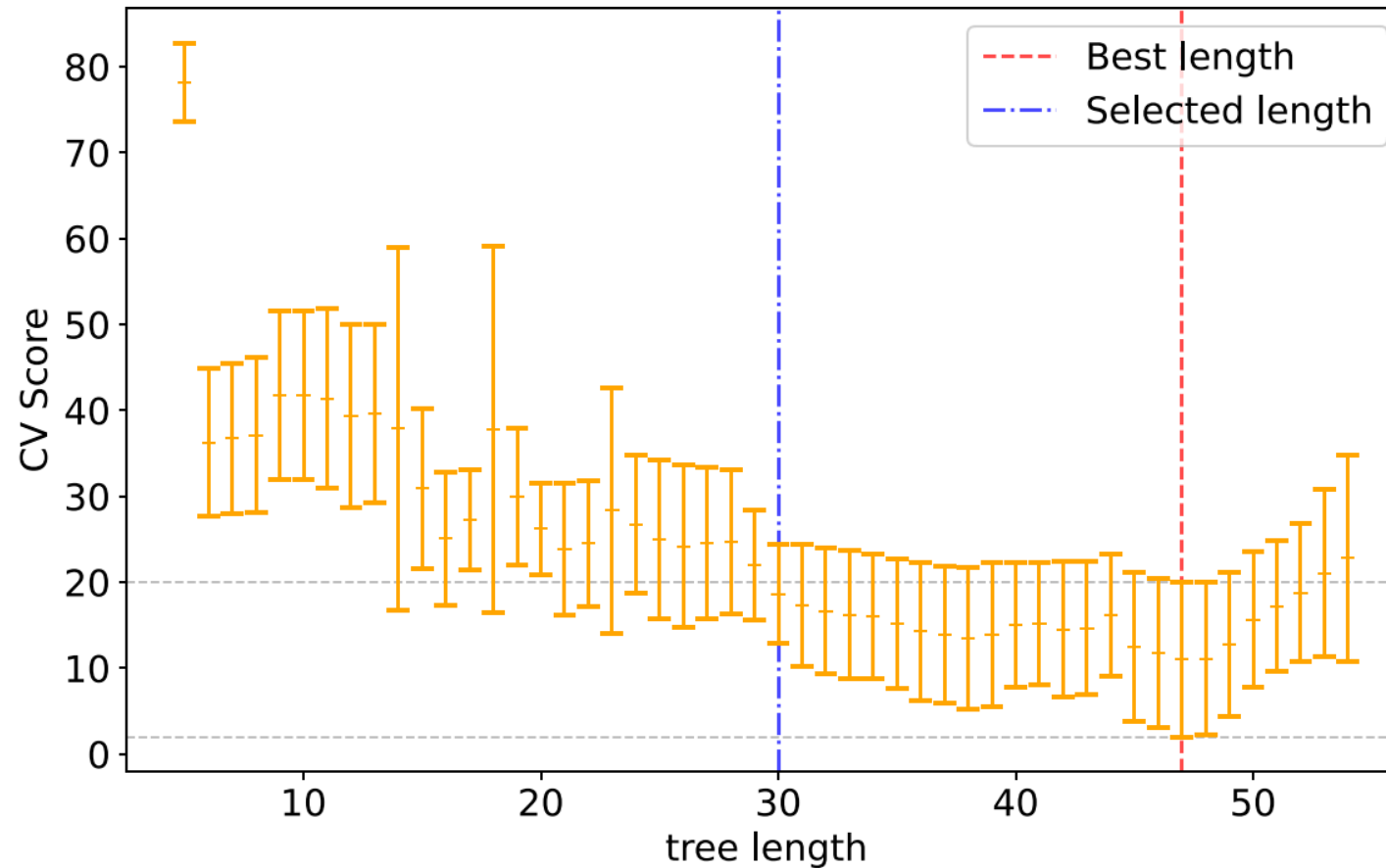


Figure 5.2: Cross validation results for yield strength.

GP Configuration

- Population size: 1000
- Generations: 100
- Crossover probability: 90%
- Mutation probability: 5% per node (for remaining 10% of individuals)
- Tournament selection size: 2
- Maximum expression length: 50
- Function set: $\{+, -, *, ^, \exp\}$
- Terminal set: $\{\text{var}, \text{parameters}\}$
- NSGA-II multi-objective optimization (NLL, DL-complexity)
- Gaussian likelihood σ_{noise} estimated
- Least-squares optimization of model parameters

UTS

$$\left(t \theta_1 + (T \theta_2)^{t \theta_3} + \theta_4\right) \left(\theta_5 \left((T \theta_6)^{t \theta_7} - e^{t \theta_8 (T \theta_9)^{T \theta_{10}}}\right) + t \theta_{11} + \theta_{12}\right) + \theta_{13}$$

UE

$$(t \theta_1 + \theta_2)^{T \theta_3} (t (T \theta_4 + \theta_5) + \theta_6) + \theta_7$$

EB

$$(t \theta_1 + \theta_2) \left((t \theta_3)^{T \theta_4 + \theta_5} \right)^{\theta_6} + \theta_7$$